



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time Table) this portion gives further specific details regarding the course.

Course No	: EEE F214/INSTR F214/ECE F214/SNS F242
Course Title	: ELECTRONIC DEVICES
Instructor-in-Charge	: NAVNEET GUPTA (ngupta@pilani.bits-pilani.ac.in) (Chamber Bo.: 2210-H)
Lecture Instructor	: Navneet Gupta and Arnab Hazra
Tutorial Instructors	: Sandeep Joshi (T1), Satyendra K Maurya (T2), Rahul Kumar (T3), Arnab Hazra (T4), Prashant Kumar (T5), Y Sujan (T6), Ritesh Vyas (T7)

1. Course Description (as per the Bulletin):

Crystal structure and growth of semiconductor, electrical conduction in solids, Elementary quantum physics (Photoelectric effect, uncertainty principle, Schrodinger wave equation and tunneling), energy bands in solids, charge carriers in semiconductors, excess carriers in semiconductors, Fabrication of p-n junctions, equilibrium conditions, forward and reverse biased junctions, metal-semiconductor junctions Bipolar junction transistors, field effect transistors (JFET, HEMT, MOSFET), Special diodes (varactor diode, solar cell, LEDs, Tunnel diode and HBT).

2. Scope and Objectives:

The course provides a comprehensive understanding of the characteristics, operation and limitations of semiconductor devices. This course starts with explanation on crystal structure, semiconductor materials, fundamentals of semiconductors. This course helps in developing the understanding about excess carriers in semiconductors and its application. In-depth study on 'junctions' prepares the students for even a detailed study on devices to be studied later like FET, BJT, MOSFET viz. commonly employed in Integrated Circuit (IC) technology. Concepts of optoelectronic diodes will also be discussed in this course.

3. Course Learning outcomes:

At the end of this course, students will be able to:

- **CLO1:** Explain the fundamentals of semiconductor materials, crystal structures, electrical conduction, quantum physics concepts, and energy band theory governing electronic devices.
- **CLO2:** Analyze charge carrier transport, excess carrier behavior, and the electrical characteristics of semiconductor materials under equilibrium and non-equilibrium conditions.
- **CLO3:** Evaluate the operation and characteristics of semiconductor junctions, including p-n junctions, metal-semiconductor junctions, and their behavior under forward and reverse bias conditions.
- **CLO4:** Explain the operating principles, characteristics, and applications of major semiconductor devices, including BJTs, JFETs, HEMTs, MOSFETs, and special semiconductor diodes.
- **CLO5:** Apply the fundamental principles of semiconductor device physics to analyze the performance and limitations of electronic and optoelectronic devices used in modern integrated circuits and electronic systems.

4. Text Book and References:

TB: B. G. Streetman, and Sanjay Banerjee, “Solid State Electronic Devices”, PHI learning Private Limited.

RB1: Donald A Neamen and D.Biswas, “Semiconductor Physics and Devices”, Tata McGraw Hill Limited.

RB2: Robert F Pierret, “Semiconductor Devices Fundamentals” Pearson Education, Inc.



5. Course Plan:

Module	Lec. No.	Descriptions	Learning Objectives	Reference to text
Semiconductor: Introduction and Basics	1-2	Brief history, recent trends and crystal lattices	To know about semiconductor and its importance in the vast majority of all solid state devices.	1.1-1.3
	3-4	Semiconductor models Charge carriers, effective mass.	To understand energy bands and how bandgaps are formed. To appreciate the idea of doping in semiconductors.	3.1-3.2
Semiconductor: Carrier Modeling	5-7	Statistics of donors and acceptors, Space charge neutrality.	To use the density of states and Fermi-Dirac statistics to calculate carrier concentration and to understand the concept of compensation.	3.3
	8	Conductivity and mobility, Hall effect	To understand the effect of electric and magnetic fields on drift of carriers	3.4-3.5
Excess carriers in Semiconductors	9-10	Optical absorption, excess carrier, Quasi-Fermi level	To understand how photons interact with semiconductors, generation recombination of excess carriers.	4.1 – 4.3
	11-13	Diffusion of carriers, built-in field, continuity equation	To introduce the diffusion of carriers, transport equation and quasi-fermi levels in non-equilibrium.	4.4
PN Junction Electrostatics	14-16	Equilibrium conditions, Fermi levels, Poisson's Equation, Depletion layer model.	To determine the band diagram of a p-n junction in equilibrium.	5.2
PN Junction Diode: IV Characteristics	17-20	Carrier injection, diode equation, and I-V characteristics, biasing, Reverse-bias breakdown	To know the current flow components in an ideal diodes	5.3-5.4
PN Junction Diode: Small-Signal & Transient Response	21-23	Capacitance of p-n junctions, Transient Analysis, non-ideal behavior.	To understand depletion capacitance due to dopant charges and diffusion capacitance due to mobile carriers.	5.5-5.6
Metal-Semiconductor Junction	24-25	Band-diagram of MS junction, Schottky diodes, & Ohmic contacts	To understand the purpose of MS junction and difference between MS and PN junction.	5.7
Field Effect Transistors	26-28	Junction Field Effect Transistors (JFETs), MESFET, HEMT & MISFET	To understand the structure and working of JFET, MESFET, HEMT and MISFET.	6.2 – 6.4
	29-33	MOS structure, MOS Transistors, I-V characteristics and secondary effects	To determine MOS C-V behavior and threshold voltage. To understand MOSFET band diagram, effective channel mobility, body effect and	6.5



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			subthreshold slopes and also to analyze second-order effects.	
Bipolar Junction Transistors	34-37	BJT operations, terminal current, Gummel plot, small-signal model, charge control model	To analyze BJT band diagrams, amplifications, terminal currents and biasing.	7.1, 7.2, 7.4 – 7.6
Optoelectronic Diodes	38-40	Photodiodes, Solar cells, and Light Emitting Diodes (LEDs) and their advanced version	To know the material aspects, working principle and applications of various optoelectronic diodes	8.1, 8.2, 10.1, 10.3 and lecture notes

6. Evaluation Scheme:

Component	Duration	Marks (300)	Date & Time	Evaluation type
Mid Sem Test	90 min	90	05/10 AN1	Open Book
Quizzes*	15 min	60	Tutorial hour	Closed Book
Class Participation**	-	20	Lecture hour	Open Book
Comprehensive Exam.	3 hrs	130	03/12 FN	Closed & Open Book

* Each quiz will be of 15 marks. Total 6 quizzes will be conducted out of which best 4 will be considered for grading. 50% quizzes will be surprise.

****Class Participation:** This component evaluates student engagement during lecture hours. This is also linked with the attendance in the lecture class.

Criterion for NC: A student should obtain 25% of the average of the class, or 10% of the course total marks, whichever is lower, to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded an NC report.

7. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

8. Chamber Consultation Hour and Notices: To be informed by respective tutorial instructor. All notices will be put up on NALANDA (LMS-e portal).

9. Make-up Policy: Make-up for mid-sem and comprehensive examination will be considered **ONLY** in genuine cases ((accompanied by proof). (**NOTE: no make-up will be given for quizzes and class participation**)).

(Prof. Navneet Gupta)
Instructor-in-charge
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